BIOSZEN User Manual (English)

This manual explains how to run BIOSZEN workflows from data import to reproducible export.

1. Prerequisites

- R >= 4.1 installed.
- BIOSZEN app launched from `App.R` or `BIOSZEN::run_app()`.
- Input files for **Load Data** can be `Excel` (`.xlsx`, `.xls`) or `CSV` (`.csv`).
- Curve files for **Load Curves** can be `Excel` (`.xlsx`, `.xls`) or `CSV` (`.csv`).

Reference templates are available in the app under **Reference input files (download)**:

- `Ejemplo_platemap_parametros.xlsx`
- `Ejemplo_curvas.xlsx`
- `Ejemplo_parametros_agrupados.xlsx`
- `Ejemplo_input_summary_mean_sd.xlsx`

2. Input Modes

2.1 Recommended mode: Platemap + Curves

Use this mode for full functionality and strongest statistical support.

1. Upload a platemap file in **Load Data**.
2. Upload a curves file in **Load Curves** (`.xlsx`, `.xls`, or `.csv`).

Platemap requirements:

- Sheet `Datos` with metadata and parameter columns (detailed below).
- Sheet `PlotSettings` with per-parameter axis settings (detailed below).

Detailed column definitions for `Datos` (platemap) sheet:

- `Well`: well identifier (for example `A1`, `B3`). This is essential to link platemap rows with the curves file.
- `Strain`: strain or biological group name.
- `Media`: condition or treatment (for example `Control`, `Drug A`).
- `BiologicalReplicate`: biological replicate ID (recommended: `1`, `2`, `3`, ...). Represents independent biological cultures/samples.
- `TechnicalReplicate`: technical replicate ID inside each biological replicate (for example `A`, `B`, `C` or `1`, `2`, `3`).
- `Replicate` (compatibility): legacy/alternate column used in some files. When present in compatible workflows, BIOSZEN treats it as the biological replicate reference.
- `Orden`: integer used to control display/export ordering of groups across plots and tables.
- Parameter columns: one or more numeric columns with analytes/metrics to plot and test (for example viability, fluorescence, `uMax`, etc.).

Practical structure notes:

- For full replicate-aware workflows and QC, provide `BiologicalReplicate` and, when available, `TechnicalReplicate`.
- The combination `Strain` + `Media` + `BiologicalReplicate` + `TechnicalReplicate` should consistently identify each experimental row.
- BIOSZEN supports common header aliases for `Strain` / `Media` (for example `Cepa` / `Muestra`, `Condicion` / `Tratamiento`).

Detailed `PlotSettings` columns:

- `Parameter`: exact parameter column name from `Datos`.
- `Y_Max`: initial upper Y-axis limit for that parameter.
- `Interval`: Y-axis tick spacing.
- `Y_Title`: Y-axis label used in the plot.

Curves requirements:

- In `Excel` (`.xlsx`, `.xls`): `Sheet1`: first column `Time`, additional columns by well (`A1`, `A2`, etc.). `Sheet2`: `X_Max`, `Interval_X`, `Y_Max`, `Interval_Y`, `X_Title`, `Y_Title`.
- In `CSV` (`.csv`): first column `Time`, additional columns by well (`A1`, `A2`, etc.). This is a single-table format (no extra sheets). Axis settings are auto-generated: `X_Max` and `Y_Max` use observed maxima, `Interval_X` and `Interval_Y` use `max/4`. `X_Title` and `Y_Title` are blank by default.

2.2 Grouped parameters mode (parameters only)

- Upload grouped-parameter workbook in **Load Data**.
- Use when you only need parameter charts and stats.
- Not valid as a curves input.

2.3 Single-file summary mode (Mean/SD/N)

- Upload summary workbook in **Load Data**.
- Parameters and curves can be detected from dedicated summary sheets.
- Useful when raw replicate rows are unavailable.

2.4 CSV mode (high-volume datasets)

- **Load Data** supports `.csv` files.
- Recommended for larger datasets because it is a lighter format.
- BIOSZEN auto-detects common delimiters (`,` , `;` , tab, or `|`).
- If CSV structure is not already platemap-ready, BIOSZEN attempts conversion to a compatible working profile.
- **Load Curves** also supports `.csv` for well-based trajectories (`Time` + well columns).
- Metadata files remain `.xlsx`.

3. Standard Workflow

1. Load primary data file.
2. Optionally load/merge curves.
3. Optionally load metadata file.
4. Choose scope (`By Strain` or `Combined`).
5. Choose plot type.
6. Apply filters and replicate selections.
7. Optionally normalize by control.
8. Run statistical analysis.
9. Add significance annotations.
10. Export plots, tables, metadata, and bundle.

4. Plot Types and How to Use Them

Boxplot

- Best for raw replicate distributions.
- Control spread with jitter, box width, and point size.
- Add significance bars or labels after stats.
- **Flip orientation (horizontal)** option swaps `X` / `Y` axes to improve readability when group labels are long.

Barplot

- Best for summarized comparison by group.
- Supports mean with error bars and optional raw points (raw mode).
- Supports manual and automatic significance annotations.
- **Flip orientation (horizontal)** option renders horizontal bars for clearer category-to-category comparison.

Violin

- Best for distribution shape with replicate overlays.
- Supports same annotation workflow as boxplot/barplot.
- **Flip orientation (horizontal)** option swaps axes to prioritize readable group/category labels.

Stacked

- Use parameter selector and parameter order controls.
- Configure deviation bars and parameter color behavior.
- Supports label or bracket significance mode.
- **Flip orientation (horizontal)** option is available for stacked views when a horizontal layout is easier to read.

Correlation

- Select X and Y parameters.
- Choose Pearson, Spearman, or Kendall.
- Optional overlays: regression line, r, p, R2, equation.
- Advanced panel supports anchor-based one-vs-all screening and Excel export.

Heatmap

- Select parameter subset.
- Scale mode: none, by row, or by column.
- Optional row/column clustering and dendrograms.
- Optional in-cell value labels.
- Cluster exports available when row clustering is enabled.

Correlation Matrix

- Multi-select parameters.
- Choose correlation method.
- Apply multiple-testing correction (Holm, FDR, Bonferroni, none).
- Optionally display only significant labels.

Curves

- Configure axis limits, labels, and line width.
- Choose line geometry and confidence interval style.
- Optional replicate trajectory display in raw mode.

Composition Panel (multi-plot layouts)

Recommended flow:

1. From each individual chart, click **Add to panel**.
2. Open the **Composition Panel** tab.
3. In **1) Plot Selection**, choose plots to combine and adjust order (move up/down, move to top/bottom, or remove).
4. In **2) Layout & Canvas**, define rows/columns, optional custom grid, column/row sizes, and final output size in px.
5. In **3) Visual Style**, configure legend mode/side, fonts, global text sizes, and palette.
6. Optional: add rich text boxes and per-plot overrides.
7. Preview and export to `PNG`, `PPTX`, or `PDF`.

Composition metadata:

- **Download metadata** stores panel layout and styling.
- **Upload composition metadata (.xlsx)** restores a composition in a new session.

5. Normalization

Enable **Normalize by control** and choose a control medium.

- BIOSZEN computes normalized columns with `_Norm` suffix.
- Correlation supports axis-specific normalization (`both` , `X only` , `Y only`).
- If strict control pairing is not possible, BIOSZEN applies fallback logic.

6. Statistics

Statistical tests and R packages (main panel)

- Shapiro-Wilk (normality): `stats::shapiro.test`
- Kolmogorov-Smirnov (normality): `stats::ks.test`
- Anderson-Darling (normality): `nortest::ad.test`
- ANOVA (significance): `stats::aov`
- Kruskal-Wallis (significance): `stats::kruskal.test`
- t-test (significance): `rstatix::t_test` and `rstatix::pairwise_t_test`
- Wilcoxon (significance): `rstatix::wilcox_test`
- Multiple-testing correction (Holm/FDR/Bonferroni): `stats::p.adjust`

Post hoc paths by selection:

- Tukey and Games-Howell: `rstatix`
- Dunn: `rstatix::dunn_test`
- Dunnett: `DescTools::DunnettTest`
- Scheffe, Conover, Nemenyi, and DSCF: `PMCMRplus`

Curve statistics (S1-S4):

- S1 (global shape difference): `stats::lm` + `splines::ns` + `stats::anova`
- S2 (pointwise Fisher comparison): `stats::pnorm` + `stats::pchisq`
- S3 (endpoint difference): `stats::pnorm`
- S4 (AUC): `gcplyr::auc` , with normality-based method selection (`stats::shapiro.test`) and comparison via `stats::t.test` , `stats::wilcox.test` , `stats::aov` , or `stats::kruskal.test` as applicable

Comparison modes:

- All vs all
- Control vs all
- Pair

Multiple-testing correction:

- Holm
- FDR
- Bonferroni
- None

Summary mode notes:

- Normality may be unavailable (`NA`).
- Some non-parametric paths requiring raw observations are disabled.

7. Significance Annotation Workflow

Manual

1. Select Group 1 and Group 2.
2. Enter annotation label (for example `*` , `**` , `***` , `ns`).
3. Add, reorder, edit, or remove annotations.

Automatic

1. Run significance tests.
2. Open auto-annotation options.
3. Choose inclusion (`significant only` or `all`).
4. Choose label mode (`stars` or `p-value`).
5. Replace or append existing annotations.

8. QC and Replicate Management

Use QC panels to review:

- Missing values.
- Outliers by group.
- Sample size and replicate coverage.

Biological replicates

- Manual include/exclude controls.
- Automatic IQR-based filtering.
- Keep-N reproducibility-oriented selection.

Keep-N automatic selection:

- Ranks replicates by reproducibility (distance to the group median across parameters).
- Keeps the closest replicates to that center (lowest score).
- Combine it with IQR outlier filtering when you want stricter replicate selection.

Technical replicates (inside each biological replicate)

- Dedicated **Technical replicate QC** tab (shown when valid technical replicates exist).
- Group and biological-replicate selectors to view and manually deselect technical replicates.
- Global buttons to select all or deselect all technical replicates.
- Automatic technical outlier detection with IQR.
- Technical Keep-N to keep the N most reproducible technical replicates per subgroup.

9. Metadata, Versioning, and Reproducibility

Metadata workflow:

- Export current UI state with **Download metadata**.
- Re-import metadata to restore compatible settings.
- The **Flip orientation (horizontal)** state is preserved in metadata export/import roundtrip.

Regression coverage:

- Automated regression tests verify that orientation flip applies only to `Boxplot`, `Barras` (Barplot), `Violin`, and `Apiladas` (Stacked).
- They also verify metadata roundtrip persistence and orientation application in final plot builders.

Versioning and bundle workflow:

- Save plot versions in-session.
- Build reproducibility ZIP with plot assets and metadata.
- Reopen work with consistent configuration.

10. Downloads

Main outputs include:

- Plot image (`PNG`, `PDF` depending on plot type).
- Data export.
- Metadata export.
- Statistics export.
- Bundle ZIP.
- Advanced correlation table export.
- Merged platemap/curves exports (when merge tools are used).

11. Growth Module

The growth tab extracts growth parameters from uploaded files:

- Accepted file type in this tab: `Excel` (`.xlsx`).
- Supported input structures (auto-detected):
  - Raw reader/Tecan-like layout (data in `Sheet1`, typically starting on later rows with leading index columns).
  - Processed table layout starting at cell `A1`: first column is time (for example `Time`) and each additional column is one curve/well.
- This means parameter extraction works even when files are not in the original instrument layout, as long as the time + curves structure is respected.

Definition of extracted GrowthRates parameters:

- `uMax`: estimated maximum slope in the exponential phase (specific growth rate).
- `max_percap_time`: mean time window where maximum per-capita growth is detected.
- `doub_time`: estimated doubling time as `ln(2) / uMax`.
- `lag_time`: estimated transition time before sustained exponential growth.
- `ODmax`: maximum measured signal/OD value in the curve.
- `max_time`: time at which `ODmax` is reached.
- `AUC`: area under the curve over the analyzed time window.

Typical use:

1. Upload one or more growth files.
2. Set max time and interval.
3. Run extraction.
4. Download ZIP output.
5. Import extracted outputs into plotting workflow when needed.

12. Troubleshooting

- **Upload error**: verify required sheets and column names.
- **No plot generated**: confirm selected strain/parameter is present after filters.
- **Normalization unavailable**: verify control medium exists in current scope.
- **Stats disabled**: check whether your input mode supports that test path.
- **Curve merge issues**: ensure platemap merge is loaded first for well remapping.
- **CSV not recognized**: validate required headers and a consistent delimiter (`data .csv` : `Strain` / `Media` ; `curves .csv` : `Time` + well columns).
- **Slow performance**: reduce selected parameters, disable heavy overlays, or filter groups.

13. Support

For support or bug reports: `bioszenf@gmail.com`